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Characterization of human-like menstruation in the spiny mouse: comparative studies with the human and induced mouse model

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STUDY QUESTION: Is the newly discovered menstruating rodent, the spiny mouse, a valid model for studying endometrial morphology and menstruation?

SUMMARY ANSWER: Our study is the first to demonstrate a primate-like pattern of natural menstruation in a rodent, with decidualization, spiral arteriole remodeling and piece-meal endometrial shedding.

WHAT IS KNOWN ALREADY: The spiny mouse has a naturally occurring menstrual cycle. This unique feature has the potential to reduce the heavy reliance on primates and provide a more appropriate small animal model for menstrual physiology research.

STUDY DESIGN, SIZE, DURATION: This study compares morphological changes in the endometrium during early, mid and late menstruation of the spiny mouse ($n = 39$), human ($n = 9$) and the induced mouse model of menstruation ($n = 17$).

PARTICIPANTS/MATERIALS, SETTING, METHODS: We assessed tissue morphology with hematoxylin and eosin and erythrocyte patterns with Mallory's trichrome. We conducted staining for neutrophil gelatinase associated lipocalin (NGAL), cytokeratin and interleukin-11 (IL-11) in all species. We used double immunofluorescence staining for vascular endothelial growth factor and alpha-smooth muscle actin to detect vasculature remodeling and western immunoblot to detect interleukin-8 (IL-8) and macrophage migration inhibitory factor (MIF) in the menstrual fluid of spiny mice.

MAIN RESULTS AND THE ROLE OF CHANCE: Menstruation occurs in the spiny mouse over a 72-h period, with heaviest menstrual breakdown occurring 24 h after initial observation of red blood cells in the vaginal cytology. During menstruation, the endometrium of the spiny mouse appeared to resemble human menstrual shedding with focal epithelial breakdown observed in conjunction with lysis of underlying stroma and detection of IL-8 and MIF in menstrual fluid. The mouse exhibits extensive decidualization prior to induced menses, with transformation of the entire uterine horn and cytokeratin expression absent until initiation of repair. Decidualization occurred spontaneously and was less marked in the spiny mouse, where epithelial integrity remained intact. In all species, the decidua was positive for IL-11 secretion and neutrophil recruitment was similar in each. Spiral arteriole formation was confirmed in the spiny mouse.

LARGE SCALE DATA: N/A

LIMITATIONS, REASONS FOR CAUTION: This is a descriptive study comparing primarily morphological traits between the spiny mouse, the mouse and the human. Reagents specific to the spiny mouse were limited and resources for global use of this novel species are few.

WIDER IMPLICATIONS OF THE FINDINGS: Our work supports the spiny mouse as a viable model, sharing many attributes of physiological menstruation with humans. The strength of a natural as opposed to an artificial model is validated through the striking similarities observed between the spiny mouse and human in uterine breakdown. The spiny mouse may be highly useful in large-scale investigations of menstruation and menstrual disorders.

STUDY FUNDING/COMPETING INTEREST(S): N.B. and S.R. are each recipients of a Research Training Program scholarship supported by Monash University. This work was supported by the Victorian Government Operational Infrastructure and laboratory funds to H. D. The authors declare no competing interests.

Key words: menstruation / spiny mouse / decidualization / animal model / inflammation

Introduction

True menstruation, the cyclical shedding of the endometrial functionalis resulting from rapid progesterone withdrawal, is a rare biological phenomenon. Menstruation occurs almost exclusively in higher order primates, including women, with several species of bats and the elephant shrew being the exceptions as the only non-primate species to menstruate (Van der Horst and Gillman, 1941; Rasweiler, 1991; Rasweiler and De Bonilla, 1992; Zhang *et al.*, 2007).

It was widely accepted that the order Rodentia does not naturally menstruate, with experimental intervention necessary to induce menstrual-like changes in laboratory rodents. The laboratory mouse (*Mus musculus*) exhibits an estrous cycle and, failing viable pregnancy, naturally resorbs endometrial tissue. In contrast, menstruating species shed the functionalis at the end of an infertile cycle. This is largely attributable to the fact that menstruating species have a spontaneous decidual reaction, in which decidualization of endometrial stromal cells occurs under the influence of high circulating levels of progesterone. This reaction occurs irrespective of signals from an implanting blastocyst, which is required for decidualization in non-menstruating species. These decidual cells cannot be de-differentiated and therefore, in the absence of luteotropic chorionic gonadotropin, luteolysis proceeds. A rapid decline of progesterone results in production of chemokines, infiltration of immune cells, protease activation and endometrial sloughing of decidua during menstruation (Evans and Salamonsen, 2014).

Menstruation induction in the mouse requires the females to be ovariectomized and the uterus to be hormonally primed using sequential estrogen injections and progesterone implants. Decidualization must then be mechanically induced. Only through artificial decidualization in conjunction with progesterone withdrawal by implant removal can the common mouse exhibit menstrual-like shedding.

We recently discovered that the common spiny mouse (*Acomys cahirinus)* undergoes cyclic endometrial shedding with frank menses observed. To our knowledge, this is the only rodent to have a natural menstrual cycle, with a new cycle initiated every 8–9 days (Bellofiore *et al.*, 2017). We propose that the spiny mouse presents a suitable small animal model of menstruation. This study directly compares the timing and nature of epithelial and stromal endometrial breakdown and repair in the induced mouse model, the spiny mouse and women to validate our assertions.

Materials and Methods

Ethics

Human ethics

All tissue collections were approved from Institutional Ethics Committees at Monash Health and Monash Surgical Private Hospital (Ethics 03066B & 06014 C).

Endometrial biopsies were collected by curettage from normally cycling women following informed consent. By this method, the functionalis and a small amount of basalis endometrium are usually sampled. All women had regular menstrual cycles (28–32 days), were under 40 years of age and had not received steroid hormone therapy in the last 6 months. All women had morphologically normal endometrium as assessed by an experienced pathologist. Menstrual cycle staging was assessed using the Noyes criteria. Tissues obtained during menstruation were used in the current study, during early (Day 1, $n = 3$), mid (Day 2, $n = 3$) or late (Day 5–7, $n = 3$) menstruation, and during the proliferative phase ($n = 3$).

Animal ethics

All experimental procedures adhered with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes. All research was approved in advance by the Monash University/Monash Medical Centre Animal Ethics Committee (MMCA 2016/40).

Husbandry: spiny mice

Virgin female spiny mice, (SpM, a total $n = 39$) aged 12–16 weeks and weighing 29–35 g, were used from our in-house colony and maintained as previously described (Bellofiore *et al.*, 2017).

Mouse model of menstruation (MMoM)

Female C57BL/6 mice (a total $n = 17$), aged 8–12 weeks, were maintained as previously described (Brasted *et al.*, 2003).

Induction of menstruation and tissue collection

Uterine tissues from the induced mouse model of menstruation (MMoM) were sourced from previous studies (Brasted *et al.*, 2003; Kaitu'u-Lino, 2007). Ovariectomized females received a subcutaneous injection of estradiol-17B (E_2) 100 ng for 3 days. After 3 days rest, a progesterone implant was inserted subcutaneously with injections of 5 ng E_2 continued. The lumen of the uterine horn was injected with sesame oil (20 µl) to stimulate decidualization. Progesterone implants were removed 48 h after decidualization induction, and mice were euthanized by carbon dioxide inhalation: at the time of implant removal ($n = 3$); after 24 h ($n = 7$) and after 48 h ($n = 7$). Uterine horns were fixed in 10% buffered formalin for 48 h, and dehydrated in 70% ethanol for 24 h, before processing to paraffin wax; tissues were embedded transversely and sectioned at 5 µm.

Spiny mouse post-mortem analysis and timed tissue collection

Female spiny mice were euthanized by isoflurane overdose at following menstrual cycle stages; proliferative ($n = 6$), secretory ($n = 7$) and menses. Menstrual collection occurred at stages determined from the time of first sign of erythrocytes in the vaginal cytology (obtained through vaginal lavage, encompassing early (0–12 h, $n = 9$), mid (24–36 h, $n = 9$) and late menstruation (48–72 h, $n = 10$), based on our previous data of a 3-day menstrual bleed. One uterine horn was fixed transversely, and one longitudinally; both were processed and sectioned as described above. Additional spiny mice ($n = 4$) were culled during mid menstruation (24 h) to collect blood and debris via saline flush into the uterine horn for western immunoblot analysis.

Staining

Tissues were stained with hematoxylin and eosin (H and E) for overall morphological comparisons and Mallory's trichrome to identify vasculature containing erythrocytes, as per previously described protocols (Bellofiore *et al.*, 2017).

Immunohistochemistry

To assess epithelial integrity and neutrophil infiltration denoting inflammatory processes across menstruation, tissues were stained for cytokeratin and neutrophil gelatinase-associated lipocalin (NGAL), respectively.

Tissue samples were dewaxed through three changes of xylene for 5 min each, rehydrated through graded ethanol for 2 min each and then submerged in dH_2O for 5 min. If relevant, antigen retrieval was then performed (Supplementary Table I). Slides were washed in Tris-Buffered Saline with Tween-20 (0.05%) (TBS-T, 2×5 min), followed by one 5 min wash in TBS, and blocked with 3% hydrogen peroxide solution for 20 min occurred prior to washing as above. Serum block was applied for 30 min prior to primary antibody incubation overnight (4°C). Negative controls substituted host species IgG for the primary antibody. Sections were washed (3×5 min) in TBS before being incubated in secondary biotinylated IgG (Vector Laboratories, CA) at 1:200 dilution for 1 h. Following washing (three times 5 min in TBS), sections were incubated in avidin-biotin complex for 45 min, then washed. 3, 3′-Diaminobenzidine (DAB) was applied for 1–3 min. Tissues were rinsed in dH_2O before counterstaining in 10% Harris hematoxylin for 3 min. Slides were dehydrated and coverslipped as above.

Immunofluorescence

To assess the changes to vasculature across the menstrual cycle in the spiny mouse, tissues were subjected to double immunofluorescent staining for vascular endothelial growth factor (VEGF) and alpha smooth muscle actin (aSMA) (Supplementary Table I).

Western immunoblots

Western immunoblot was used to determine the presence of inflammation and repair factors in the menstrual fluid of spiny mice during mid menses (24 h). Menstrual flushings, containing sloughed endometrial tissue, were homogenized in radioimmunoprecipitation assay (RIPA) buffer prior to refrigerated centrifugation for 10 min. Supernatant was retained and 4× SDS loading buffer containing B-mercaptoethanol was added. Lysates were denatured at 95°C for 5 min with 15 µl of sample run on a 10% SDS polyacrylamide gel. Membranes were blocked in 5% skim milk for 90 min prior incubation in primary antibodies, interleukin-8 and macrophage migration inhibitory factor (IL-8, R&D Systems #AF-208-NA and MIF, R&D systems, #AF-289-PB), overnight at 4°C. Immunoblots were washed prior to incubation for 90 min in anti-rabbit horseradish peroxidase conjugated antibody (1:1000, Dako #D0487), followed by development using Clarity western enhanced chemiluminescence substrate (BioRad, #1705061), and visualized using the BioRad ChemiDoc MP Imaging System. Membranes were stripped and reprobed for B-actin loading control (1:10,000) as above.

Image acquisition

Immunofluorescent images were captured using a Nikon C1 Confocal microscope, and all other images using an Olympus BX41 bright-field microscope. To analyze DAB stained tissues, three randomly selected fields of view (FOV) per uterine sample were taken at ×200 magnification. These images were blindly analyzed using ImageJ software (National Institutes of Health) that provided an automated analysis of area coverage (%) of immunopositive cells (denoted by brown/black staining). The average of these FOV were calculated to give a mean value per sample which were then compared across species and stage of menstrual cycle. Samples were analyzed as follows: For Hu, $n = 3$/group for all stains; SpM, $n = 6–9$/group (NGAL and cytokeratin) and $n = 3$/group for IL-11; MMoM $n = 3–7$/group for all stains.

Statistical analysis

All statistical analyses were conducted using GraphPad Prism Version 6.01 (GraphPad Prism Software Inc. CA). Non-normally distributed data is presented as median (25th percentile, 75th percentile) and normally distributed data is presented as mean ± standard deviation (SD). Significant differences were calculated using Kruskal–Wallis test or ANOVA for non-normal or normal data, respectively.

Results

Timing of menstrual breakdown in the spiny mouse

Overall, mean body and uterine weight of female spiny mice were 32.38 ± 2.68 g and 0.074 ± 0.029 g, respectively, with no differences observed at any timepoint during menstruation at the time of post-mortem. A positive correlation between female body weight and uterine weight was observed ($r = 0.42$, $P = 0.03$, Pearson correlation). We found no evidence that female weight in this cohort had any impact on whether we observed frank menses, and no significant relationship was observed between body weight and length of menstrual period (Fig. 1A,B). From the histology, 54% of spiny mice fell within the expected timing of menstruation based on our previous data of an average 3-day menstrual bleed. Slight deviations within 24 h of expected menstruation were noted. A total of 21% exhibited endometrial breakdown slightly prior to (7%) or slightly later than (14%) expected, while 25% exhibited substantially longer (7%) or substantially shorter (18%) bleeding times (>24 h discrepancy) than previously recorded (Fig. 1C).

Median body weight of females that had noticeably shorter menses was less than those with longer menses (Fig. 1C). However, neither

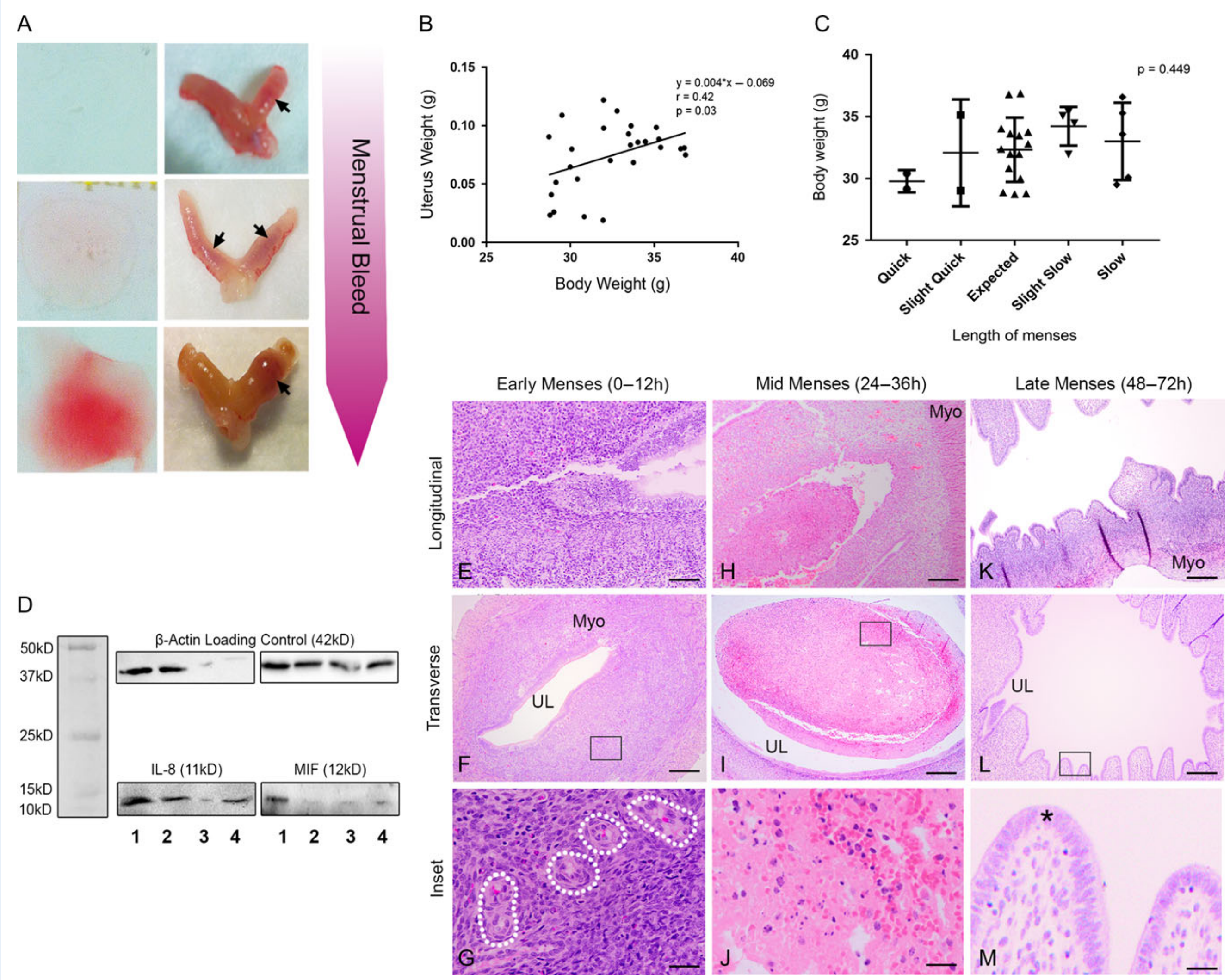

Figure 1 Menstrual breakdown in the spiny mouse. Myo = myometrium, L = luminal space (**A**) Variation in the degree of bleeding between individual animals is observed, ranging from light to heavy, visible in both vaginal lavage and corresponding uterine dissection. (**B**) Uterine weight is positively correlated with body weight (linear regression, $P = 0.03$). (**C**) Correlation between body weight and expected length of menstruation, where 'slight quick/slow' are within 24 h of expected histological point of menstruation, and 'quick/slow' are more than 24 h of histological deviation from expected point of menstruation. No significant relationship is observed. (**D**) Western blot of individual, heavily bleeding spiny mice (1–4) during mid menses (24 h). Inflammation markers interleukin-8 (IL-8) and macrophage migration inhibitory factor (MIF) are shown to varying degrees in all samples, however are most prominently detected in 3/4 and 2/4 female spiny mice respectively. (**E**) At 0–12 h following initial bleeding, the stroma of the decidualized endometrium begins to break away. The thickness of the differentiated endometrium renders the luminal space relatively small (**F**). Spiral arterioles can be identified at higher magnification (**G**, dotted lines). Between 24–36 h in both longitudinal and transverse sections, endometrial sloughing (**H**, **I**) of decidua (**J**) peaks, with menstrual debris in the lumen (L). Menstrual breakdown nears completion 48–72 h after initiation of bleeding, with the majority of endometrium shed, a clear and larger lumen (**K**, **L**), and regenerated luminal epithelium (*) (**M**). Scale bars (E, F, H, I, K, L) = 200 μM and (G, J, M) = 50 μM.

body nor uterine weights were a significant contributing factor for the differences observed.

Western blot analysis confirmed the presence of both IL-8 and MIF protein in the menstrual debris of heavily bleeding females at variable levels. IL-8 was most obvious in 3 of 4 samples, and MIF was in 2 of 4 samples (Fig. 1D).

Up to 12 h after first evidence of red blood cells in vaginal cytology, the beginning of the degradation of decidualized endometrial stroma containing spiral arterioles (Fig. 1E–G) was observed. Transverse sections showed that decidualization does not cause a complete closure of the uterine lumen. During mid-menstruation (Fig. 1H–J), 24–36 h after initial observation of cytological bleeding, large masses of decidualized tissue are shed in discrete regions of the uterine horn. The later stages of menstruation (Fig. 1K–M), between 48 and 72 h, shows near completion of the shedding and regeneration process, with a visibly larger luminal space and reduced thickness of the endometrium. The luminal epithelium has regenerated (*) with the cessation of peeling, and fewer fragments of old epithelial debris are present in the lumen.

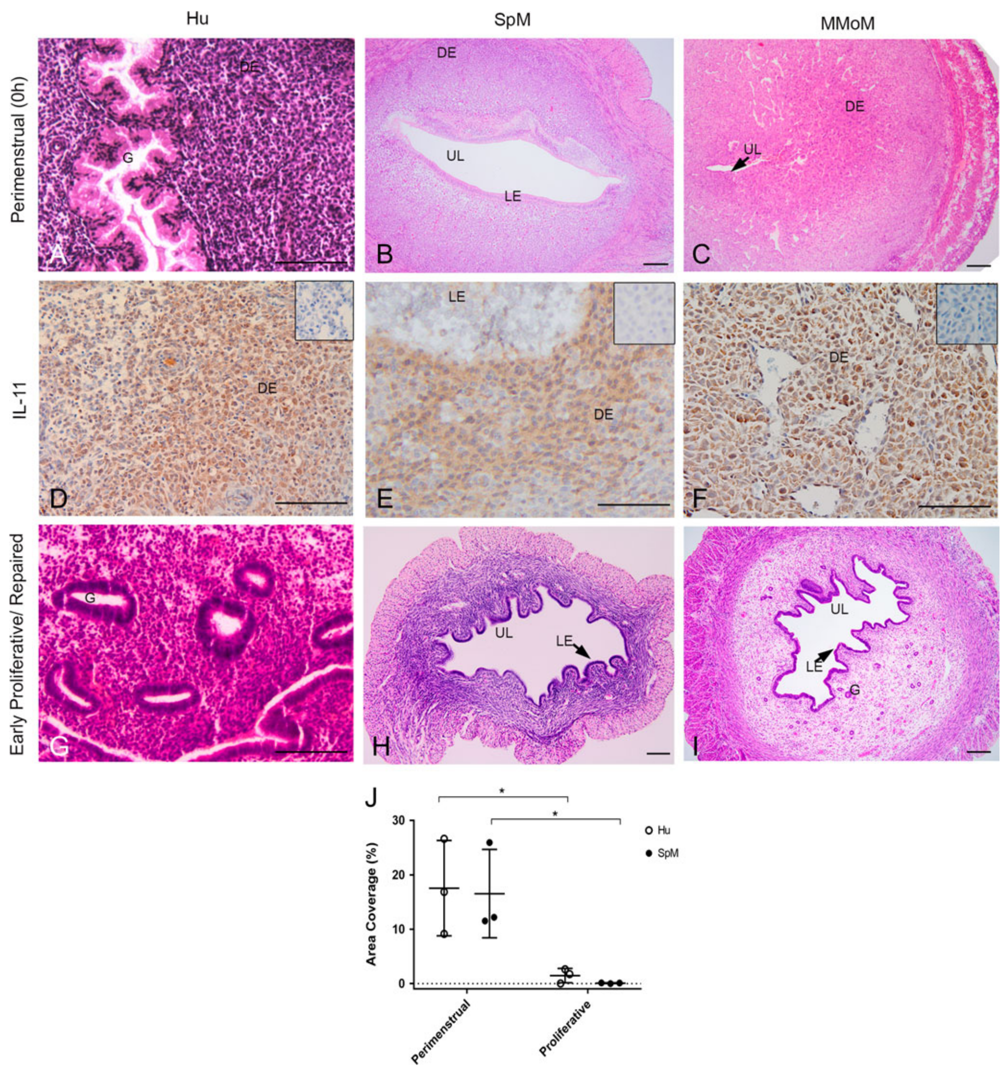

Figure 2 Comparative decidual reaction and resultant morphology of the human (Hu), spiny mouse (SpM) and mouse model of menstruation (MMoM). Myo = myometrium, UL = uterine lumen, LE = luminal epithelium and DE = decidualized endometrium. Early menstrual sections of the uterine horn (~0 h) stained with H and E in the human (functionalis only) (**A**), SpM and MMoM (**B**, **C** respectively, whole uterus, transverse plane). Tubular secretory glands (G) are prominent in the human. MMoM exhibits an extreme decidualized endometrium which has caused almost complete loss of the lumen and luminal epithelium. Myometrial structures are condensed at the periphery with histological artefacts (tearing/holes) in the musculature. SpM maintains the lumen, epithelium, and an intact bilayer of the myometrium. All species exhibit immunopositive brown staining for IL-11 (**D**–**F**) in the decidualized endometrium. Samples of corresponding negative controls are depicted in inset squares. Proliferative endometrium shows new glands in the proliferative human endometrium and repaired (48 h) MMoM with uterine lumen now evident in MMoM and proliferative SpM tissues (**G**–**I**). Area (%) of decidualized endometrium is significantly decreased after shedding in Hu and SpM (**J**). Scale bar = 100 μM.

Comparative endometrial morphology between species

All species undergo a decidual reaction (Fig. 2A–C). Secretory glands are pronounced in human early menstrual endometrium. Transverse sections of MMoM at the time of implant removal (0 h), show an extensive decidual reaction encompassing most of the endometrium, and obstruction of the lumen by highly differentiated stroma. The myometrial structures are separating and friable (Myo). In contrast, the spiny mouse exhibits a less extensive decidual response, with the lumen and luminal epithelium (LE) intact and readily observable. A distinguishable myometrial bi-layer is clear in SpM, but is not in MMoM.

Decidualized stroma in all species express IL-11 (2D–F), with staining most pronounced during the latest stages of the secretory phase, immediately prior to initiation of menstrual breakdown. Decidualization is significantly reduced ($P < 0.01$, Two-way ANOVA) after entering the proliferative phase where all decidual tissue has been shed in human and SpM tissues, nor is it evident in the repaired (48 h) endometrium of MMoM (Fig. 2G–J, Supplementary Fig. 1). Invariably, the decidualized endometrium in MMoM is entirely transformed at the time of implant removal (0 h).

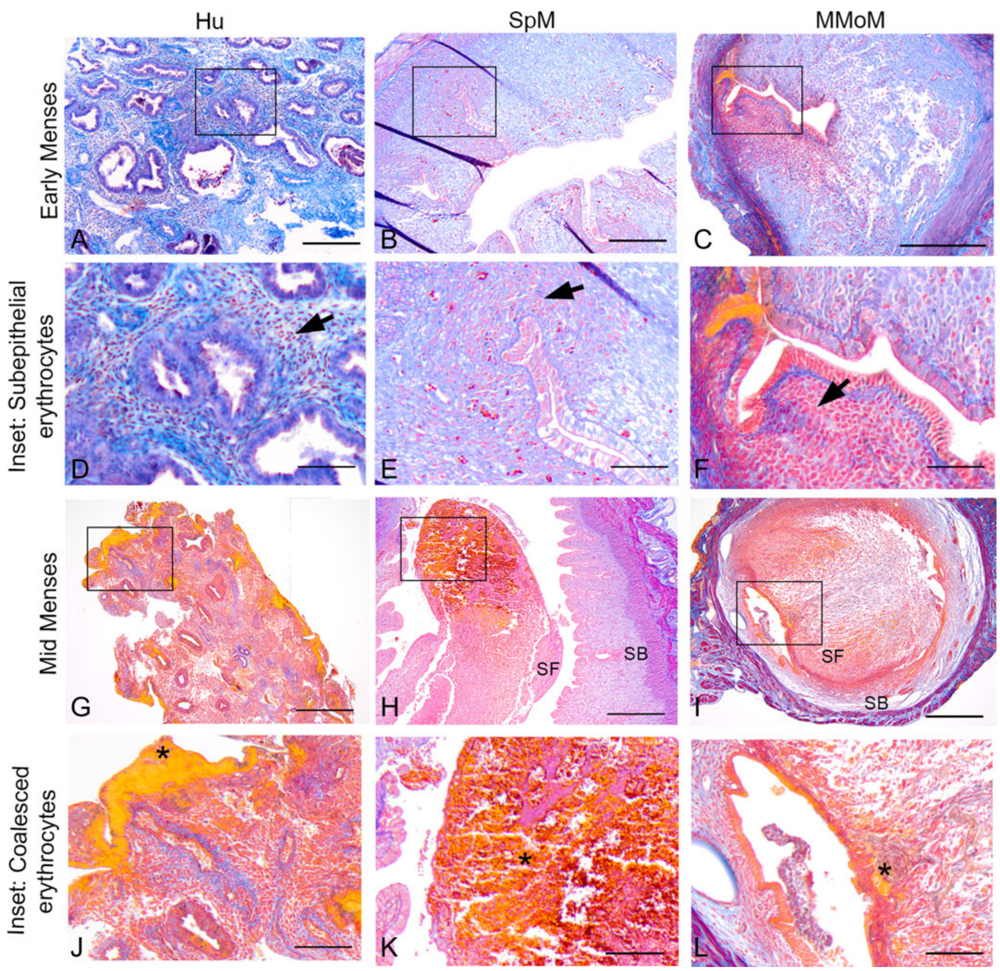

Figure 3 Comparative early and mid-menstrual phase endometrium in the human (Hu), spiny mouse (SpM) and mouse model of menstruation (MMoM). Mallory's trichrome stain. In all three species, sub-epithelial bleeding occurs during the early initiation of stromal breakdown (0–12 h) (**A**–**C**), denoted by regions of erythrocyte infiltration (arrows) (**D**–**F**). Coalescing of erythrocytes is pronounced during mid menstruation (Hu 24–48 h, MMoM and SpM 24 h) (**G**–**I**). Pools of red blood cells (**J**–**L**) are easily recognized in the human and spiny mouse and, to a lesser extent, in MMoM (*). Shedding persists into the stratum basalis (SB) layer in MMoM until the myometrium is reached, but not in SpM, which is restricted to the stratum functionalis (SF). Scale bar (A–C, G–I) = 200 µM and (D–F, J–L) = 50 µM.

During early menstruation in all three species (Fig. 3A–F), patches of sub-epithelial red blood cell infiltration (arrows) are distinguishable. During mid-menses (3G–L), bright orange pools of erythrocytes are notable in all species (*). The integrity of the stratum basalis (SB) in MMoM is compromised during the shedding process, as evidenced by fragmentation and tearing away of the SB from the stratum functionalis (SF). In SpM, shedding is limited to the SF.

Endometrial inflammation, breakdown and repair during menstruation

Cytokeratin positive staining in the early menstrual stages in luminal endometrial epithelium is absent in MMoM. The luminal space is swallowed by extensive decidualization, and epithelial structures are not readily identified. In contrast, both the human and spiny mouse show notable cytokeratin staining in the luminal epithelium (Fig. 4A–C). Progressive menstrual breakdown reduces (but does not abolish) this signal in Hu and SpM (Fig. 4D–F, arrows), with little expression in MMoM until repair initiates. The return of a strong cytokeratin signal coincides with epithelial repair during the later phases of menstruation across all three species (Fig. 4G–I). MMoM demonstrate active menstrual shedding by 24 h with centralized cellular debris present in the lumen (Fig. 4F), although variation exists between females. Some demonstrate continued closure of the lumen with the majority decidua remaining intact and the beginning of SF tearing away (Fig. 3I) at 24 h. SpM shows higher area fraction of cytokeratin positive endometrium during repair than both Hu and MMoM (J-K, Two-way ANOVA, $P < 0.05$). SpM exhibits distinctive regions of focal shedding (*), whereby unshed regions of decidua, distinguished from newly generated stroma by its compact and cobblestone appearance of tightly adhered cells (#), lie adjacent to regions of matrix degradation and epithelial peeling.

NGAL expression in the human and spiny mouse endometrial glands at the initiation of menstruation is similar (Fig. 5). Neutrophils are present in the endometrial stroma, but are mostly discernible in the myometrial

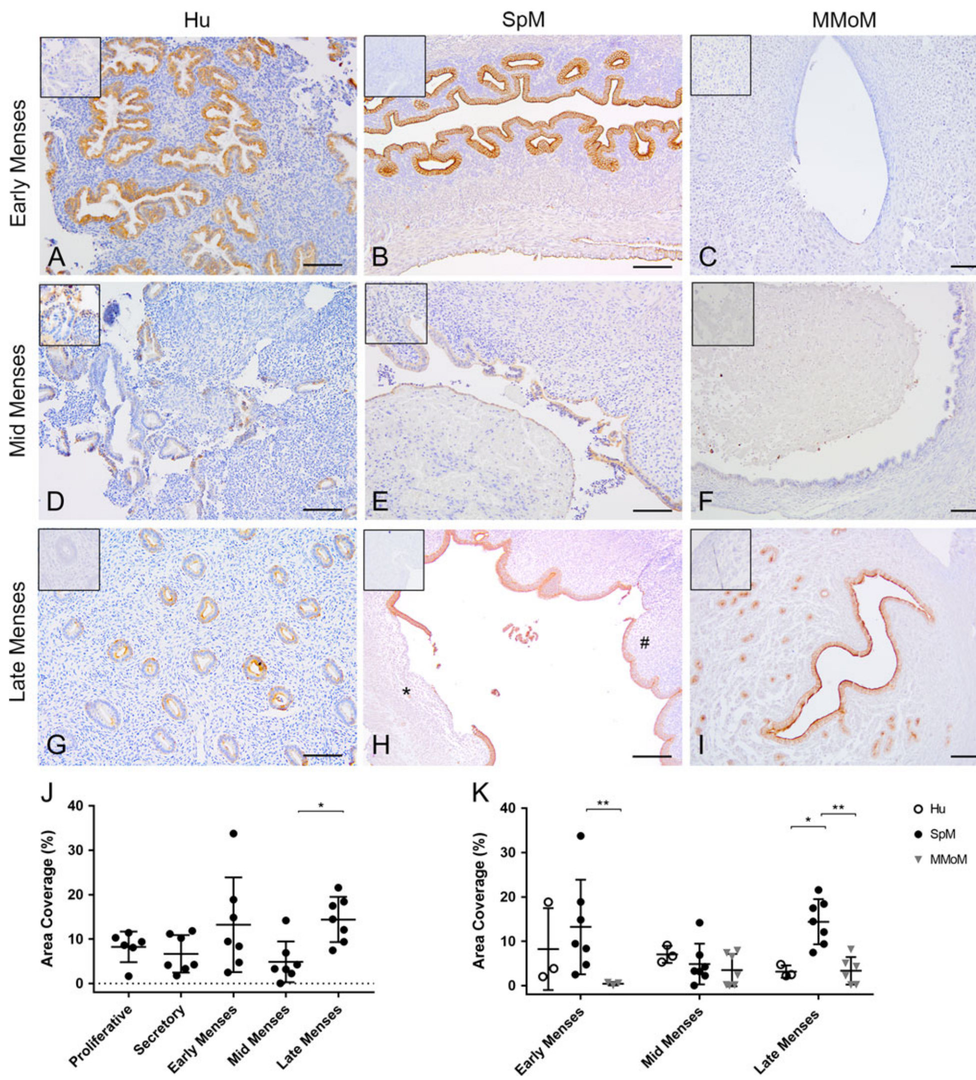

Figure 4 Comparative epithelial changes during menstruation in women (Hu), the spiny mouse (SpM) and the mouse model of menstruation (MMoM). During early menses, stromal and luminal glands are relatively intact in both Hu and SpM (**A**, **B**), but no cytokeratin positive staining is apparent in MMoM (**C**). Cytokeratin signals are minimal during sloughing in all species (**D**–**F**) during mid-menses. As endometrial repair occurs, a strong cytokeratin signal returns in the final phases of menstruation across all species (**G**–**I**). Distinct regions of focal shedding (*) are observed underneath peeling luminal epithelium, opposite to yet unshed regions of compacted, decidualized stroma (#). Scale bars = 100 μM. Samples of corresponding negative controls are depicted in inset squares. A significant decrease in cytokeratin expression is demonstrated in the spiny mouse during mid menses (breakdown) compared to late menses (repair) (**J**, ANOVA, $P < 0.05$). Compared to SpM, cytokeratin expression in MMoM is significantly reduced in early and late menses, and significantly reduced in Hu during late menses (**K**, Two-way ANOVA, $P < 0.001$).

tissue of the spiny mouse. Sporadic clusters of neutrophils are evident in the mouse in the endometrium bordering the myometrium during early menstrual breakdown. Few neutrophils are evident in Hu during early menses (Fig. 5A–C). As menstruation progresses, neutrophil infiltration peaks across all three species (Fig. 5D–F) prior to a marked reduction nearing the end of menstrual breakdown (Fig. 5G–I). A significant increase ($P = 0.01$, ANOVA) in mean area coverage of NGAL immunopositive staining is seen in the functionalis of the spiny mouse uterus during mid menses compared to the proliferative phase (Fig. 5K), and compared to early menses in Hu and MMoM (Fig. 5J–K).

Spiral arteriole remodeling in the spiny mouse endometrium

During the late secretory phase of the spiny mouse menstrual cycle (Fig. 6), spiral artery assemblies in the endometrium (Endo, *) are immunopositive for aSMA (green), while the endothelium and surrounding stroma show positive expression of VEGF (red, arrows, Fig. 6A–F). VEGF staining localizes to the luminal epithelial cells and uterine fluid, with reduced staining in the endometrium during the proliferative phase (arrows, Fig. 6G–I). VEGF is present in the myometrium (Myo) in both phases.

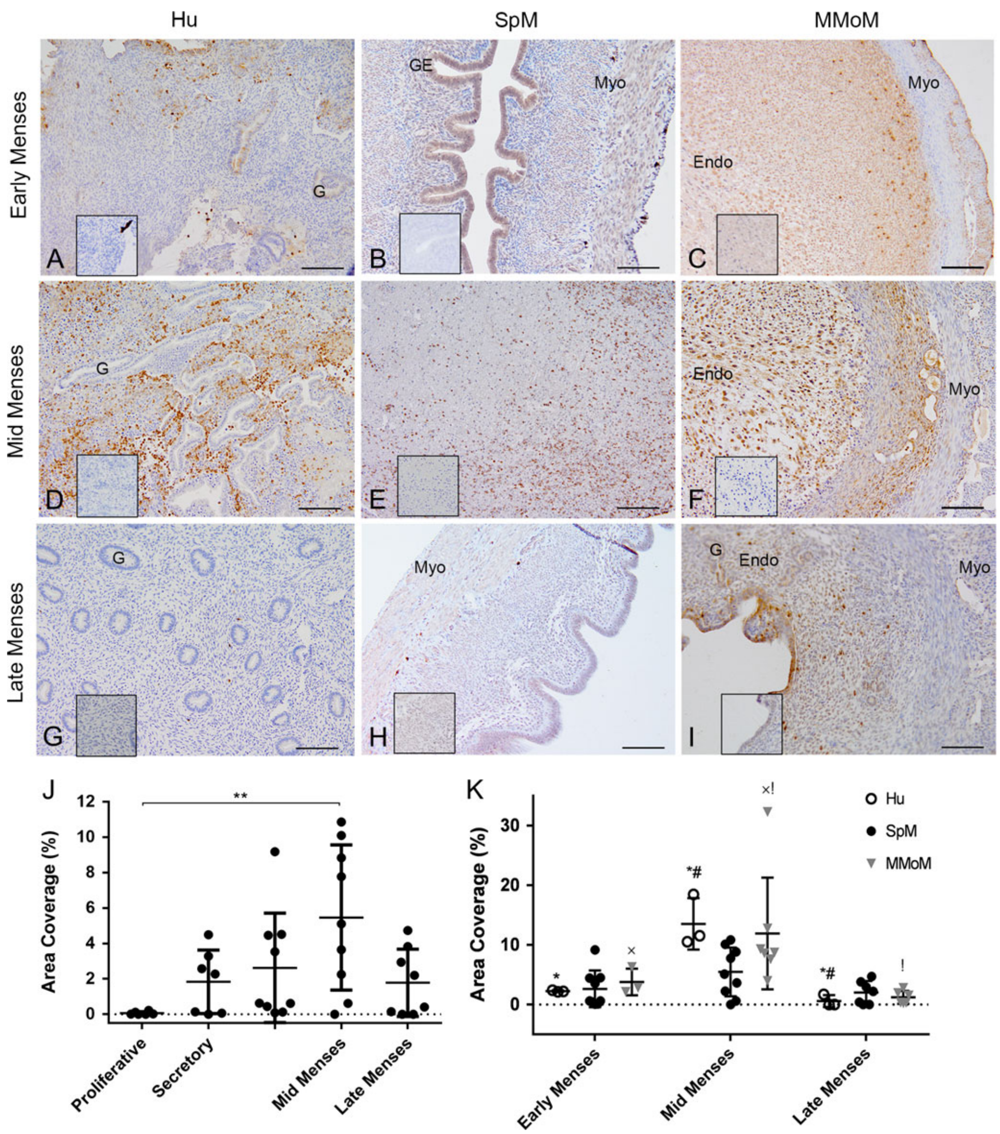

Figure 5 Comparative changes to NGAL expression and neutrophil recruitment during menstruation in women (Hu), the spiny mouse (SpM) and the mouse model of menstruation (MMoM). Neutrophils have not yet obviously infiltrated the endometrial stroma. Glandular epithelial NGAL staining is seen in Hu and SpM. In SpM, neutrophils appear clustered in the myometrium. (**A–C**). During mid menses, neutrophilic influx is at its greatest in all species (**D**–**F**). During the final stages of menstruation, neutrophils are distinctly reduced (**G**–**I**). A significant increase in NGAL in the functionalis is seen during the mid menstrual phase compared to the proliferative phase in the spiny mouse (J, $P < 0.01$, ANOVA). No differences in NGAL expression are seen between species, however all show an increase during mid menses (K, $P < 0.001$, Two-way ANOVA). Like-symbols denote groups which are significantly different from each other within same species (*, # and!). Scale bar = 200 μM. Samples of corresponding negative controls are depicted in inset squares.

A comparison of features of menstruation between the human, mouse and spiny mouse is summarized in Table I.

Discussion

Our data agrees with our initial observation of an average 3-day menstrual bleed in the spiny mouse, with 75% of females examined menstruating according to the range of time previously described (Bellofiore et al. 2017). In the spiny mouse, the heaviest endometrial degradation occurs ~24 h after initial observation of red blood cells in the vaginal cytology. The process of menstrual shedding in this species is less predictable than that observed in MMoM, which exhibits menstrual-like bleeding and resolution of repair within 48 h after removal of progesterone (Kaitu'u-Lino et al., 2007). However, this menstruation event is controlled by the removal of exogenous hormone mediated support rather than endogenous reduction in hormone levels upon corpus luteum demise, which will clearly lead to variation in onset of bleeding. The importance of the spiny mouse menstrual model is that it exhibits natural variation between individuals in the overall length of the menstrual cycle. The application of the spiny

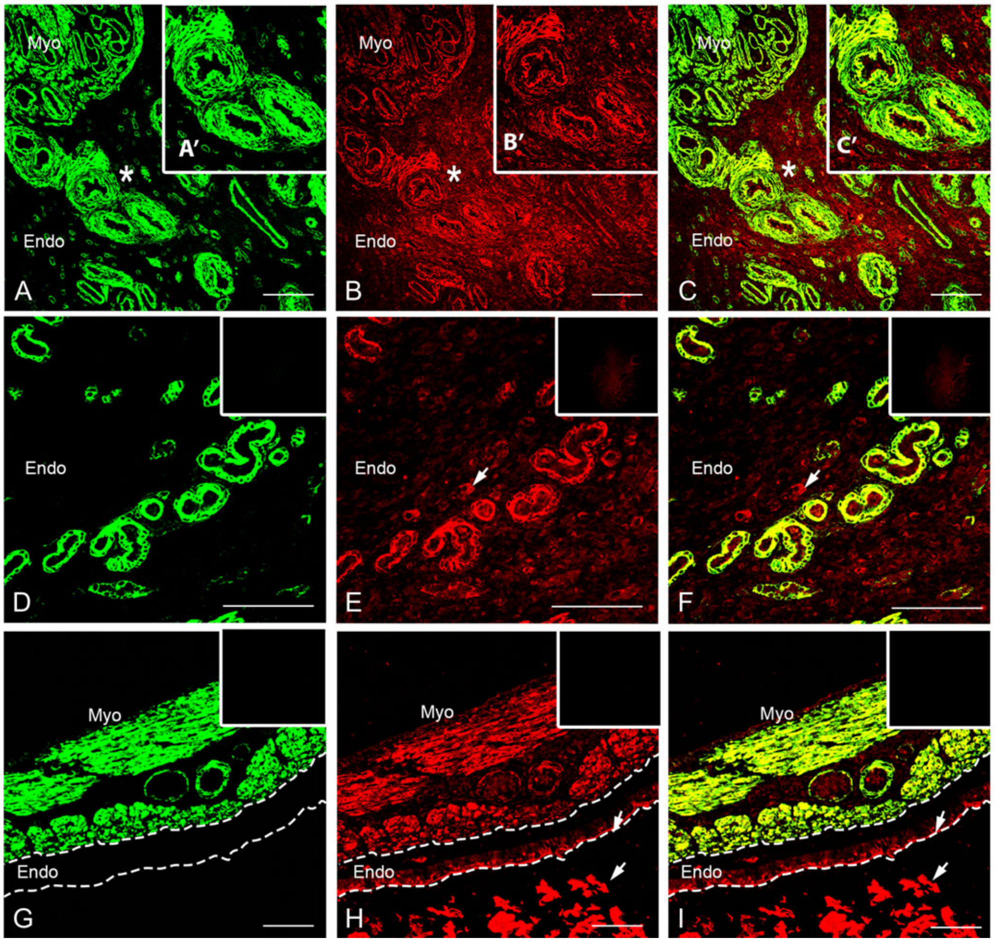

Figure 6 Spiral arteriole formation in the endometrium of the spiny mouse. Green = aSMA, red = VEGF and yellow = merge. **A–F**: Late secretory phase. Clusters of aSMA positive spiral arteries with VEGF positive endothelium (*) stemming from the myometrial base (Myo) of the uterus into the endometrium (Endo) (**A–C**). Inset square shows magnified spiral arteries (**A', B', C'**). aSMA (**D**), VEGF (**E**) and merge (**F**) positive spiral arterioles penetrating endometrium (Endo) at higher magnification. VEGF also stains positively in endometrium stromal cells (**E–F**, arrows). **G–I**: Mid proliferative phase of the spiny mouse menstrual cycle. aSMA positive spirals are not present in the endometrium (**G**). VEGF is positive in the myometrium (Myo), endometrial luminal epithelium and uterine fluid (arrows) (**H**).

mouse as a model for abnormal uterine bleeding, including infrequent and heavy menstrual bleeding, presents an attractive opportunity for further research.

While the uterus of the spiny mouse anatomically corresponds to the Rodentia order in possessing a bicornuate uterus, the physiological likeness to humans of the spontaneous cyclic decidualization observed in the spiny mouse overcomes the pitfalls of using an artificially induced animal model. Although the human uterine samples demonstrate only functionalis as opposed to MMoM and SpM, the extensive and uncontrolled endometrial decidualization observed in MMoM does not recapitulate the human endometrium prior to menses. The extent of this decidual reaction forces not only the endometrial glands and myometrial bilayer to the periphery of the uterine horn but also compresses the lumen and luminal epithelium, and is resolved only after repair is complete. In contrast, the spiny mouse shows natural thickening and receding of the endometrium, while the structural integrity of the organ in its entirety is not compromised. A lumen lined by an intact epithelium is evident overlying decidualized stromal cells.

IL-11 has been proven a classical marker of decidua in humans during the peri-implantation phase (Dimitriadis *et al*., 2000) and presence of IL-11 immunoreactivity in the decidualized spiny mouse uterus reinforces this is a true decidual event.

The spiny mouse endometrium lacks distinctive remodeling of the glands from elongated (proliferative phase) to coiled (secretory phase), as we see in human endometrial tissue. We do observe invaginations of the luminal epithelium, and a few glands protruding from this region, but not to the same degree as seen in women. It may be possible that in the spiny mouse, luminal epithelial glands alone can facilitate the necessary secretions during the secretory phase of the menstrual cycle. Further studies are required in this area.

Humans and spiny mice exhibit similar timing and degree of bleeding relative to the length of the menstrual cycle in each species. All three species are similar in the initiation of menstruation, with focal regions of degrading stromal matrix and infiltration of erythrocytes, and are again comparable in terms of their heaviest day of menstrual. However, the unusual extent of shedding, likely to be due to the

Table I Summary comparison of menstruation in the human (hu), spiny mouse (SpM) and Mouse model of menstruation (MMoM).

	Hu	SpM	MMoM
Reproductive tract	Simplex uterus with fundal body Fallopian tubes	Bicornuate uterus with uterine horns Oviducts	Bicornuate uterus with uterine horns Oviducts
Menstrual cycle	Natural	Natural	Artificially induced
Shedding	Menstrual waves	Menstrual waves	Menstrual waves
Length of menses	5–7 days	2–4 days	1–3 days
Decidualization (Peak % area of endometrium)	Spontaneous IL-11 positive 17.54 ± 8.77	Spontaneous IL-11 positive 16.54 ± 8.13	Artificially induced IL-11 positive 11.75 ± 1.30
Cytokeratin signal in luminal epithelium (% Area of endometrium at early menses)	Present throughout $8.25 + 9.27$	Present throughout $13.23 + 10.67$	Minimal until menstrual repair initiates 0.45 ± 0.36
Peak neutrophil influx (% Area of endometrium)	Mid menses (48 h) 13.52 ± 4.31	Mid menses (24 h) 5.47 ± 4.10	Mid menses (24 h) 11.91 ± 9.37
Peak erythrocyte infiltration	Mid menses (48 h)	Mid menses (24 h)	Mid menses (24 h)
Spiral artery remodeling	Yes	Yes	No
Glands	Both epithelial and intrauterine	Mostly epithelial	Both epithelial and intrauterine

extensive mass of the decidua that develops in the induced model, appears to cause regions of the stratum basilis to undergo shedding in conjunction with the functionalis. In contrast, the basalis in the spiny mouse remains intact, and is more reflective of menstruation in women, in which only the upper two-thirds of the endometrium is shed (Ferenczy, 1980).

We demonstrated further structural similarities between the spiny mouse and human through analysis of epithelial changes across menstruation. The persistence of cytokeratin staining throughout the menstrual process in the spiny mouse is comparable to the re-epithelialization and repair which occurs in women prior to the cessation of menstrual breakdown, ensuring the breakdown and repair process is simultaneous (Ferenczy, 1980; Nair and Taylor, 2010). In this study, the luminal epithelium of the MMoM was not immunopositive for cytokeratin during early menses. This is in contrast to previous studies which identified new proliferating luminal epithelial cells lining the residual basal stroma within 8 h of bleeding onset (Cousins, *et al.*, 2014). In many of the MMoM sections, early menses tissues contained little to no uterine lumen. This may be the consequence of luminal damage during administration of the decidual stimulus (oil), however, more likely reflects the extensive decidual reaction which appears to severely compress the luminal epithelial cells, and consequently shows significantly less cytokeratin staining than SpM. Possibly due to both (a) small sample size and (b) the wave like nature of menstruation, and thus, random sampling of incomplete repaired tissue, we did not observe a significant increase in cytokeratin expression during late menses in Hu or MMoM.

Transverse sections of the uterus are traditionally used for histological analysis in reproductive studies of rodents. However, we demonstrate that longitudinal sections during menstruation show focal regions of simultaneous matrix degradation and epithelial shedding adjacent to unshed decidua in SpM. This pattern of smaller localized shedding and neighboring areas of endometrium yet to undergo breakdown supports the menstrual waves theory demonstrated in both the MMoM and women (Kaitu'u-Lino, 2007; Garry *et al.*, 2009), where not all endometrial tissue is shed at once, but rather in fragmentary ripples to avoid mass tissue and blood loss, and allow for the repair process to take place. This method of collection likely accounts for the variability seen at 24 h in MMoM, where some sections demonstrate a total degradation of stromal matrix with shedding decidua and prominent blood loss, and others have most of the functionalis intact with less erythrocyte infiltration into the lumen.

The detection of IL-8 in menstrual debris of the spiny mouse demonstrates a typical inflammatory response during uterine shedding. This observation, in conjunction with NGAL analysis, also supports, as previous finding have in human tissues, that IL-8 is involved in attracting and activating neutrophils during menstruation (Arici *et al.*, 1998). The maximum area of coverage of neutrophils during the inflammatory process of menstruation in the spiny mouse is less than that in humans and MMoM, but not significantly so. Our findings confirm the role of neutrophils in stromal matrix destruction seen in both humans and the mouse model (Armstrong *et al.*, 2017). The presence of the pro-inflammatory and antibacterial cytokine MIF in the spiny mouse menstrual debris suggests macrophage activation, and supports recent findings that macrophages play a crucial role in both the degradation and regeneration processes of menstruation (Calandra and Thierry, 2003; Garry *et al.*, 2010; Evans *et al.*, 2016). The variation in MIF and IL-8 proteins as identified by western immunoblotting likely reflects the different amount of menstrual debris recovered from the endometrial lumen of each individual mouse. As highlighted in Fig. 1, the timing and degree of shedding at 24 h varies between spiny mice, as observed in humans, therefore the amount of shed material also varies, again, as in humans (Evans *et al.*, 2014). However, we do demonstrate that proteins identified in human menstrual fluid are also readily detectable in menstrual debris obtained from the spiny mouse, reinforcing the utility of this model as a human like model of menstruation.

An additional advantage of the spiny mouse as a natural model of menstruation is that spiral arteriole remodeling during the premenstrual phase is characteristic of menstruating primates, and cannot be induced in MMoM. Epithelial VEGF expression during the mid-proliferative phase concurs with previous findings that VEGF plays an essential role in post-menstrual re-epithelialization and repair, and detection in the stromal cells during the secretory phase is consistent with past studies of the regulative role in endometrial vascularization (Shifren *et al.*, 1996; Fan *et al.*, 2008). This suggests that the spiny mouse is a may be used to investigate disorders of poor angiogenesis and how disruption of the remodeling of this vasculature impacts pregnancy outcomes (e.g. pre-eclampsia).

We acknowledge two primary methods of euthanasia used in this study in the rodents: carbon dioxide (mouse) and isoflurane overdose (spiny mouse). Standard operating protocols for our spiny mouse colony preference isoflurane overdose given its more immediate effects due to the fragile nature of the skin (Seifert *et al.*, 2012) and highly active behavior of this species, minimizing the potential stress (Wong *et al.*, 2013) and subsequent risk of the spiny mice of sustaining injury. While mice produce corticosterone, spiny mice, like primates, secrete cortisol as their stress hormone. For the purposes of this study, we therefore maintain that the method of euthanasia could have potential implications in comparing inflammation reactions between the rodent species, but unlikely between the spiny mouse and human. Variable stress levels between the spiny mice and mouse may have altered corticosteroid production and subsequent levels, but are unlikely to significantly affect endometrial expression of immune marker NGAL given the abrupt exposure to hypoxic environment.

High throughput uses and the ability to analyze entire organs histologically are undeniably advantageous using small, non-primate species for menstrual research. Our research demonstrates the assets of the spiny mouse, in particular for vascular studies, decidualization, stromal breakdown and epithelial shedding. However, MMoM demonstrates a closer immune profile in neutrophilic recruitment and may be preferable when examining inflammatory processes and initiation of repair.

While our study is the first to demonstrate primate-like qualities of the menstrual cycle in a rodent species, and highlights the advantages of the spiny mouse for menstrual research, we acknowledge the current limitations of this model. The uniqueness of the physiological profile of the spiny mouse limits use of commercially available anti-mouse antibodies for immediate successful immunohistochemical staining. Indeed, we determined that anti-human antibodies generally provided positive immunohistochemical staining within the spiny mouse endometrium, although this may not be the case in other tissues. However, many anti-human antibodies are raised in mice, and often lead to substantial cross-reactivity in the spiny mouse. For these reasons in the current study, we were unable to optimize several markers of immune cells within the endometrium of the spiny mouse, including CD45 (total leukocytic recruitment) and CD68 (macrophages), due to either non-specific staining in the stromal matrix, and therefore inaccurate quantification, or no identifiable staining. Our colony of spiny mice may have a unique CD profile, or immune antigens which require non-standard fixation methods and/or use of other media (e.g. paraformaldehyde). Researchers must bear in mind the current lack of widespread use of spiny mice and difficulties implementing common techniques on reproductive tissues.

Our findings of the nature of menstruation in the spiny mouse, including controlled spontaneous decidualization, presence of spiral arteriole formation and, importantly, the natural variation and degree of bleeding are encouraging. We continue to support the spiny mouse as a future potential alternative, and readily available animal model, to study menstrual health and disorders.

Supplementary data

Supplementary data are available at *Human Reproduction* online.

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Authors' roles

N.B. and J.E. conceived the project, performed experiments and analyzed data. S.R. performed experiments. H.D. and P.T-S. provided critical revisions of project design and data and manuscript editing.

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Conflict of interest

The authors declare no competing interests.

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